# Proof by Exhaustion

Exam code: 9MA0

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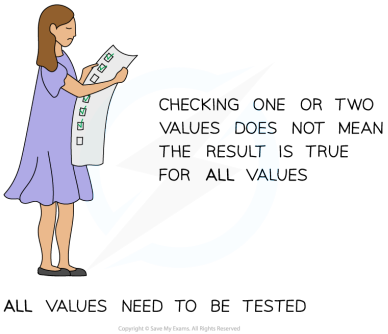
## Proof by exhaustion

### What is proof by exhaustion?

- Proof by exhaustion is a way to show that the desired result works for every allowed value

### How do I prove a result by exhaustion?

- Using proof by exhaustion means testing every allowed value not just showing a few examples

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### Difficulties with proof by exhaustion

every multiple of 4

- Aim to minimise the work involved
  - Proving a number is prime only requires testing factors up to the square root

## Examiner Tips and Tricks

- Try a simpler case if you are stuck.
- For example, if you are asked to prove that 97 is a prime number you could try thinking about what you would do for smaller primes such as 7 or 11.

## Worked Example

Prove that the difference between $n^3 + 7$ and a multiple of 7 is always 1, for all integers in the interval $1 \leq n \leq 4$.

NOTE: THIS RESULT MAY BE TRUE FOR ALL VALUES OF n BUT YOU ARE ONLY ASKED TO PROVE IT FOR INTEGERS IN THE INTERVAL $1 \leqslant n \leqslant 4$, i.e. n=1, n=2, n=3, n=4 SO 'EXHAUST' ALL POSSIBILITIES

$n = 1$, $n^3 + 7 = 1^3 + 7 = 8$ DIFFERENCE OF 1 FROM 7
$n = 2$, $n^3 + 7 = 2^3 + 7 = 15$ DIFFERENCE OF 1 FROM 14
$n = 3$, $n^3 + 7 = 3^3 + 7 = 34$ DIFFERENCE OF 1 FROM 35
$n = 4$, $n^3 + 7 = 4^3 + 7 = 71$ DIFFERENCE OF 1 FROM 70

BY EXHAUSTION, $n^3+7$ HAS A DIFFERENCE OF 1 FROM A MULTIPLE OF 7 FOR ALL INTEGERS $1 \leqslant n \leqslant 4$

DO SHOW THE APPROPRIATE MULTIPLES OF 7

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